

# SMART UTM — PROGRESS REPORT

Jönköping University · School of Engineering — status to 9 September 2026 · automated tensile testing with camera-based strain

## 18 of 20 features rig-validated; the strain channel matches a certified extensometer.

Two sessions in the materials-testing laboratory closed the validation: on the XT-205's own frames the rig's DIC agrees to 0.07 % in scale. PLA, PETG and TPU are characterised, the moisture and carbon-footprint studies are written up, and the project report is drafted. Five roadmap items remain, none of them code.

|                                                                   |                                             |                                                   |                                                             |                                                            |
|-------------------------------------------------------------------|---------------------------------------------|---------------------------------------------------|-------------------------------------------------------------|------------------------------------------------------------|
| <b>18 / 20</b><br>features rig-validated<br>(+2 verified offline) | <b>6 / 6</b><br>test protocols<br>validated | <b>30</b><br>specimens tested,<br>36 indexed runs | <b>0.99928</b><br>scale vs the XT-205<br>on the same frames | <b>5</b><br>open roadmap items —<br>rig time and one email |
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### WHERE THE PROJECT STANDS

The rig is no longer a manual machine, and its strain channel is no longer unproven. A test is: pick a recipe, follow the wizard, press Start — the software runs the protocol, watches the specimen through the camera, stops itself at fracture and writes a linked record the post-processor can re-measure. The channel is validated against itself (noise 11.9  $\mu\epsilon$ , scaling as predicted), against the literature (three polymers in band) and against a certified XT-205 video extensometer — within 1 % on yield and fracture strain, 0.07 % in scale on the same frames.

**The rig — built in-house, every control loop closed in software**



Two steppers (1.85 Nm, 20:1) drive 5 mm lead screws against a fixed base. An ANYLOAD 3 t cell measures force; a Basler camera measures strain optically at 20.8 px/mm from two markers on the 80 mm gauge. The ESP32 runs STEP/DIR/ENABLE only and knows nothing of test protocols — every protocol, guard and measurement here lives in the PyQt6 app, closing the loop at ~11 Hz.

### WHAT HAS BEEN DELIVERED

|           |                                                                                                                                                                                                                                                                            |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| June      | <ul style="list-style-type: none"><li><b>100 % infill quintet, n = 5</b><br/>UTS 46.2 <math>\pm</math> 1.2 MPa, scatter 2.5 % — publication-grade repeatability.</li></ul>                                                                                                 |
| August    | <ul style="list-style-type: none"><li><b>All six protocols validated</b><br/>Including both destructive protocols; two landed 0.9 % apart on ultimate strength, though on different specimens.</li></ul>                                                                   |
| August    | <ul style="list-style-type: none"><li><b>Fatigue and creep resolved</b><br/>Residual strength after cycling is –6.8 % against the same-protocol virgin run; creep resolved at 41<math>\times</math> the noise floor.</li></ul>                                             |
| August    | <ul style="list-style-type: none"><li><b>Laboratory cross-validation, two sessions</b><br/>80 mm and 45 mm gauge on the Zwick/Roell Z100 + XT-205; the laboratory's footage re-measured by the rig's own post-processor: scale 0.99928, R<sup>2</sup> 0.9999964.</li></ul> |
| August    | <ul style="list-style-type: none"><li><b>PETG and TPU, black PLA, the 45 mm gauge</b><br/>Three polymers in their bands; the Otsu-vs-fixed threshold question settled; noise scaling 1.78<math>\times</math> predicted, 1.78<math>\times</math> measured.</li></ul>        |
| September | <ul style="list-style-type: none"><li><b>SF17–SF20, moisture and footprint studies, report</b><br/>Capture, post-processing, noise capture and manual blob selection; the humidity and PCF sections; the 393-slide diary and the project report drafted.</li></ul>         |

### RESULTS WORTH REPORTING

**The strain channel is validated against a certified instrument**

On the XT-205's own frames: scale 0.99928, R<sup>2</sup> 0.9999964, fracture strain 5.21 vs 5.22 %, a constant –55  $\mu\epsilon$  offset (0.069 px) that vanishes from every slope. On separate specimens: yield +1.0 %, fracture strain –0.8 %. The –7.7 % UTS gap is pull rate (2.21 $\times$ ), grips and batch.

**The rig ranks materials correctly across two decades of modulus**

PLA 46.2 MPa / 3.17 GPa, PETG 41.8 MPa / 1.81 GPa, TPU 0.024 GPa to 18.9 % strain unbroken — all inside their published bands. The 50 % infill knockdown is 2.4, a matched pair repeats to 0.2 %.

**Strength repeatability is publication-grade**

100 % infill, n = 5: UTS 46.2  $\pm$  1.2 MPa — 2.5 % scatter, and the only n > 2 result we own. Two fracture protocols landed 0.9 % apart, but on DIFFERENT specimens (n = 1 each), so that bounds protocol and specimen scatter together, not the protocol alone.

**The camera is not a nicety — it changes the answer**

Over the same cycles the crosshead reads 4 % STIFFER while the specimen's true modulus falls 26 %. Without strain measured on the specimen itself, this test reports the opposite of what actually happened.

**Fatigue and creep, each resolved at n = 1**

Residual strength –6.8 % after 16 cycles at 79 % of fracture load; creep +1080  $\mu\epsilon$  at 41 $\times$  the noise floor with the rig's own drift subtracted first.

### DECISIONS I NEED FROM YOU

**The laboratory's load or crosshead channel — one email**

With either channel the strain comparison can be made at matched force, which cancels compliance and turns it into a pure scale check. **One request to Jacob Steggo at the Department of Materials and Manufacturing.**

**Rig time and a drying step for the moisture study**

The literature predicts –8.5 % strength after 24 h of ambient exposure; the rig resolves better than 1 % on a matched pair. **Three conditioned sets from one spool, one session, and a filament dryer.**

**Poisson's ratio — needs optics, not code**

The four-marker maths is written and self-tested, but the gauge is too narrow to resolve a transverse pair at the present 20.8 px/mm. **A gauge-zoomed lens (or a second camera) plus a matte backdrop unblocks true stress and Poisson's ratio.** Until then every stress reported is engineering stress, which is stated on every figure.

**More specimens before publication**

Creep and fatigue are each n = 1 at one condition. The measurement is proven; the material constants are not. **3–4 stress levels for creep and 3 repeats for fatigue** would make both publishable.

**NEXT**

**Now One pull at 2 mm/min**

One specimen, one run, no new code — separates the rate effect from specimen scatter in the –7.7 % UTS gap.

**Now Rig sessions for SF19 and SF20**

Noise capture and manual blob selection are built and verified offline; each needs one real pull to be carded as rig-validated.

**Next The controlled moisture experiment**

Dried / 24 h / one week from one spool, pulled in one session — prices the batch term and closes the laboratory comparison.